

Midterm Exam Equation Sheet

ISE 390, Fall 2026, The University of Alabama in Huntsville

1 Statistical Summaries

$$\bar{x} = \frac{1}{n} \sum^n x_i$$

$$s^2 = \frac{1}{n} \sum^n (x_i - \bar{x})^2$$

$$s^2 = \frac{1}{n-1} \sum^n (x_i - \bar{x})^2$$

$$s^2(\text{alt}) = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right)$$

$$s = \sqrt{s^2}$$

2 Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\text{Permutations of } k \text{ from } n \text{ objects: } \frac{n!}{(n-k)!}$$

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\text{Combinations of } k \text{ from } n \text{ objects: } \frac{n!}{k!(n-k)!}$$

3 Random Variables

$$\mu_x = \sum_x xp(x)$$

$$\mu_x = \int_{-\infty}^{\infty} xf(x)dx$$

$$\bar{X} = \frac{1}{n}X_1 + \dots + \frac{1}{n}X_n$$

$$\mu_{\bar{X}} = \mu$$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx$$

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

$$\text{Cov}(X, Y) = \mu_{(X-\mu_X)(Y-\mu_Y)} = \mu_{XY} - \mu_X\mu_Y$$

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}$$

$$\mu_{c_1X_1+c_2X_2+\dots+c_nX_n} = c_1\mu_{X_1} + c_2\mu_{X_2} + \dots + c_n\mu_{X_n}$$

$$\sigma_{c_1X_1+c_2X_2+\dots+c_nX_n}^2 = c_1^2\sigma_{X_1}^2 + \dots + c_n^2\sigma_{X_n}^2 - 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n c_i c_j \text{Cov}(X_i, X_j)$$

$$\sigma_{c_1X_1+c_2X_2+\dots+c_nX_n}^2 = c_1^2\sigma_{X_1}^2 + \dots + c_n^2\sigma_{X_n}^2$$

$$\sigma_x^2 = \sum_{i=1}^n p_i(x_i - \mu_x)^2 = \sum_x (x^2 p_x) - \mu^2$$

$$\sigma_x^2 = \int_{-\infty}^{\infty} f(x)(x - \mu_x)^2$$

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

$$F(x) = \int_{-\infty}^x f(t)dt$$

$$\mu_{h(X)} = \int_{-\infty}^{\infty} h(x)f(x)dx$$

$$\mu_{Y|X=x} = \int_{-\infty}^{\infty} yf_{Y|X}(y|x)dy$$

$$f(x, y) = f_X(x)f_Y(y)$$

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

4 Measurement Error

$$c_{\text{best}} = \frac{\sigma_A^2}{\sigma_A^2 + \sigma_B^2} \text{ and } 1 - c_{\text{best}} = \frac{\sigma_B^2}{\sigma_A^2 + \sigma_B^2}$$

$$\sigma_{c_1X_1+\dots+c_nX_n} \leq |c_1|\sigma_{X_1} + \dots + |c_n|\sigma_{X_n}$$

$$\sigma_U \approx \left| \frac{dU}{dX} \right| \sigma_X$$

$$\frac{\sigma_A}{A} = \sigma_{\ln A}$$

$$\sigma_U \approx \sqrt{\left(\frac{\partial U}{\partial X_1}\right)^2 \sigma_{X_1}^2 + \left(\frac{\partial U}{\partial X_2}\right)^2 \sigma_{X_2}^2 + \cdots + \left(\frac{\partial U}{\partial X_n}\right)^2 \sigma_{X_n}^2}$$

$$\sigma_U \leq \left|\frac{\partial U}{\partial X_1}\right| \sigma_{X_1} + \left|\frac{\partial U}{\partial X_2}\right| \sigma_{X_2} + \cdots + \left|\frac{\partial U}{\partial X_n}\right| \sigma_{X_n}$$